

Signals & System

Lecture 2

Signals Concepts & Properties

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Signals Concepts & Properties

- (1) Systems, **signals**, mathematical models.
Continuous-time and discrete-time signals.
Energy and power signals. Linear systems.
Examples for use throughout the course, introduction to Matlab and Simulink tools

Specific objectives for this lecture include

- General properties of signals
- Energy and power for continuous & discrete-time signals
- Signal transformations
- Specific signal types
- Representing signals in Matlab and Simulink

Lecture 2: Resources

- SaS, O&W, Sections 1.1-1.4
- SaS, H&vV, Sections 1.4-1.9
- Mastering Matlab 6
- Mastering Simulink 4

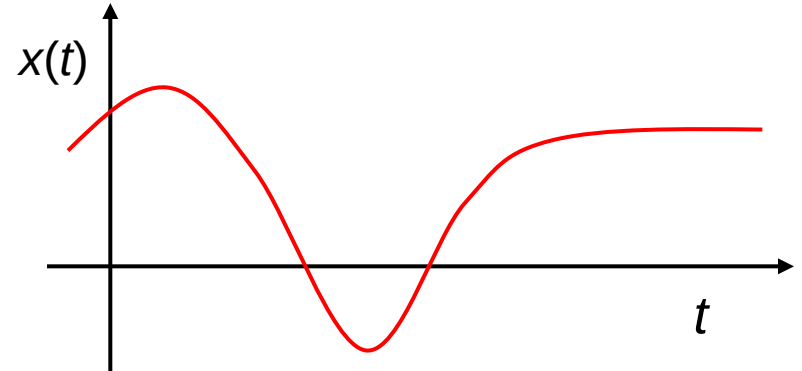
Reminder: Continuous & Discrete Signals

Continuous-Time Signals

Most signals in the real world are continuous time, as the scale is infinitesimally fine.

E.g. voltage, velocity,

Denote by $x(t)$, where the time interval may be bounded (finite) or infinite

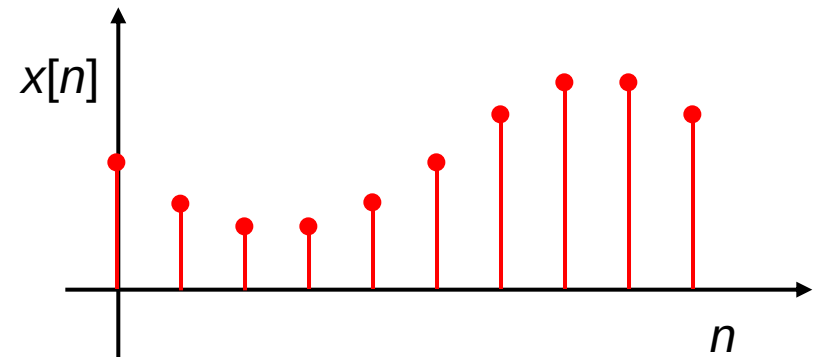


Discrete-Time Signals

Some real world and many digital signals are discrete time, as they are sampled

E.g. pixels, daily stock price (anything that a digital computer processes)

Denote by $x[n]$, where n is an integer value that varies discretely



Sampled continuous signal

$$x[n] = x(nk)$$

“Electrical” Signal Energy & Power

It is often useful to characterise signals by measures such as **energy** and **power**

For example, the **instantaneous power** of a resistor is:

$$p(t) = v(t)i(t) = \frac{1}{R} v^2(t)$$

and the **total energy** expended over the interval $[t_1, t_2]$ is:

$$\int_{t_1}^{t_2} p(t)dt = \int_{t_1}^{t_2} \frac{1}{R} v^2(t)dt$$

and the **average energy** is:

$$\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} p(t)dt = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \frac{1}{R} v^2(t)dt$$

How are these concepts defined for any continuous or discrete time signal?

Generic Signal Energy and Power

Total energy of a continuous signal $x(t)$ over $[t_1, t_2]$ is:

$$E = \int_{t_1}^{t_2} |x(t)|^2 dt$$

where $|\cdot|$ denote the magnitude of the (complex) number.

Similarly for a discrete time signal $x[n]$ over $[n_1, n_2]$:

$$E = \sum_{n=n_1}^{n_2} |x[n]|^2$$

By dividing the quantities by $(t_2 - t_1)$ and $(n_2 - n_1 + 1)$, respectively, gives the **average power**, P

Note that these are similar to the electrical analogies (voltage), but they are different, both value and dimension.

Energy and Power over Infinite Time

For many signals, we're interested in examining the power and energy over an infinite time interval $(-\infty, \infty)$. These quantities are therefore defined by:

$$E_{\infty} = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$E_{\infty} = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

If the sums or integrals do not converge, the energy of such a signal is infinite

$$P_{\infty} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$P_{\infty} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Two important (sub)classes of signals

1. Finite total energy (and therefore zero average power)
2. Finite average power (and therefore infinite total energy)

Signal analysis over infinite time, all depends on the “tails” (limiting behaviour)

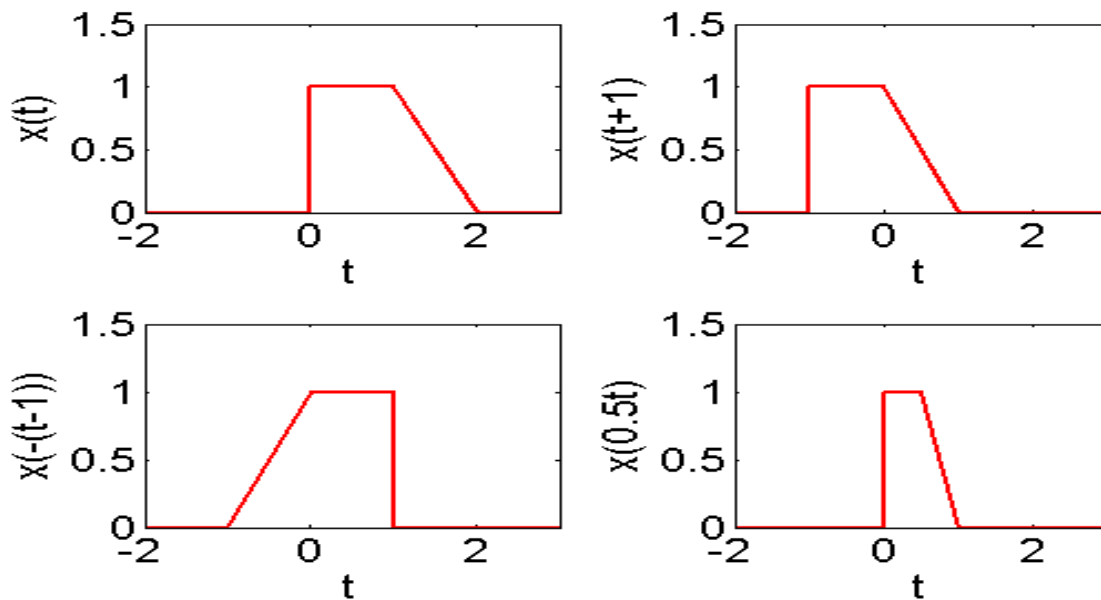
Time Shift Signal Transformations

A central concept in signal analysis is the transformation of one signal into another signal. Of particular interest are simple transformations that involve a transformation of the time axis only.

A linear **time shift** signal transformation is given by:

$$y(t) = x(at + b)$$

where b represents a signal offset from 0, and the a parameter represents a signal stretching if $|a| > 1$, compression if $0 < |a| < 1$ and a reflection if $a < 0$.



Periodic Signals

An important class of signals is the class of periodic signals. A periodic signal is a continuous time signal $x(t)$, that has the property

$$x(t) = x(t + T)$$

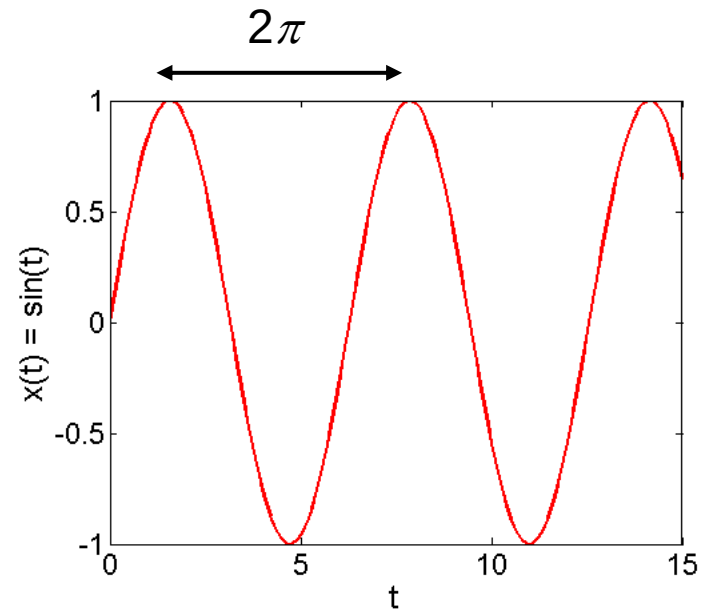
where $T > 0$, for all t .

Examples:

$$\cos(t + 2\pi) = \cos(t)$$

$$\sin(t + 2\pi) = \sin(t)$$

Are both periodic with period 2π



NB for a signal to be periodic, the relationship must hold for all t .

Odd and Even Signals

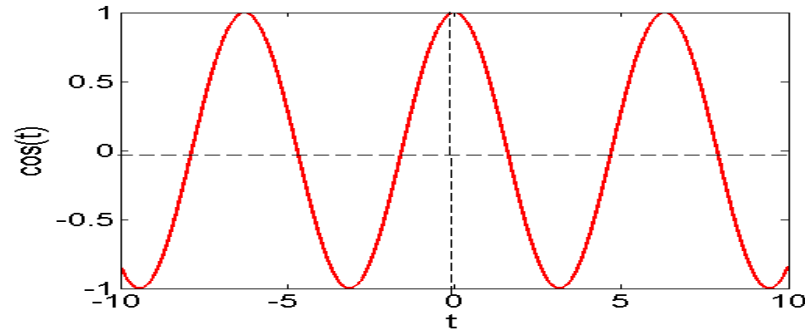
An **even** signal is identical to its time reversed signal, i.e. it can be reflected in the origin and is equal to the original:

$$x(-t) = x(t)$$

Examples:

$$x(t) = \cos(t)$$

$$x(t) = c$$



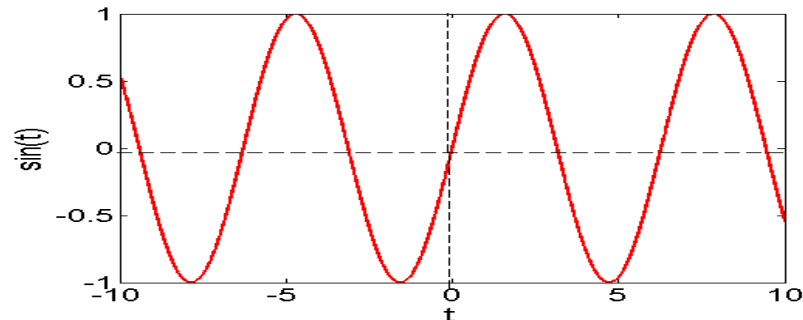
An **odd** signal is identical to its negated, time reversed signal, i.e. it is equal to the negative reflected signal

$$x(-t) = -x(t)$$

Examples:

$$x(t) = \sin(t)$$

$$x(t) = t$$



This is important because any signal can be expressed as the sum of an odd signal and an even signal.

Exponential and Sinusoidal Signals

Exponential and sinusoidal signals are characteristic of real-world signals and also form a basis (a building block) for other signals.

A generic **complex exponential signal** is of the form:

$$x(t) = Ce^{at}$$

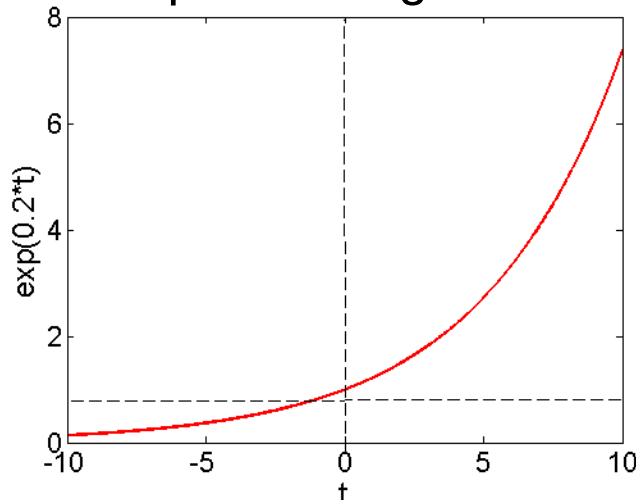
where C and a are, in general, complex numbers. Let's investigate some special cases of this signal

Real exponential signals

Exponential growth

$$a > 0$$

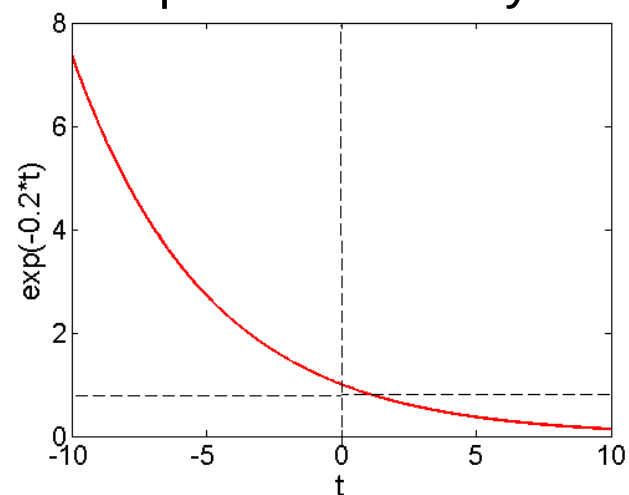
$$C > 0$$



Exponential decay

$$a < 0$$

$$C > 0$$



Periodic Complex Exponential & Sinusoidal Signals

Consider when a is purely imaginary:

$$x(t) = Ce^{j\omega_0 t}$$

By Euler's relationship, this can be expressed as:

$$e^{j\omega_0 t} = \cos \omega_0 t + j \sin \omega_0 t$$

This is a periodic signals because:

$$\begin{aligned} e^{j\omega_0(t+T)} &= \cos \omega_0(t+T) + j \sin \omega_0(t+T) \\ &= \cos \omega_0 t + j \sin \omega_0 t = e^{j\omega_0 t} \end{aligned}$$

when $T = 2\pi/\omega_0$

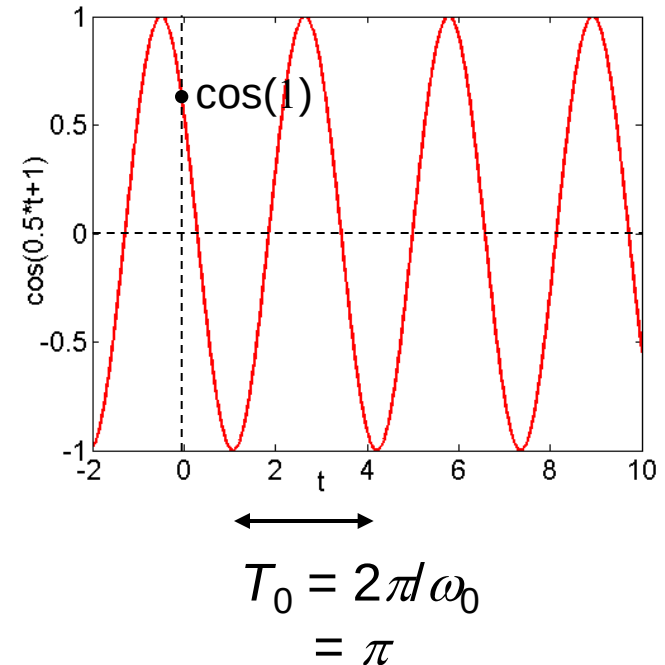
A closely related signal is the **sinusoidal signal**:

$$x(t) = \cos(\omega_0 t + \phi) \quad \omega_0 = 2\pi f_0$$

We can always use:

$$A \cos(\omega_0 t + \phi) = A \Re(e^{j(\omega_0 t + \phi)})$$

$$A \sin(\omega_0 t + \phi) = A \Im(e^{j(\omega_0 t + \phi)})$$



T_0 is the fundamental time period
 ω_0 is the fundamental frequency

Exponential & Sinusoidal Signal Properties

Periodic signals, in particular complex periodic and sinusoidal signals, have infinite total energy but finite average power.

Consider energy over one period:

$$\begin{aligned} E_{period} &= \int_0^{T_0} |e^{j\omega_0 t}|^2 dt \\ &= \int_0^{T_0} 1 dt = T_0 \end{aligned}$$

Therefore:

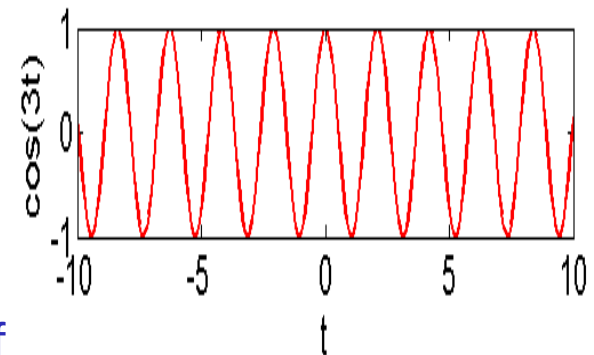
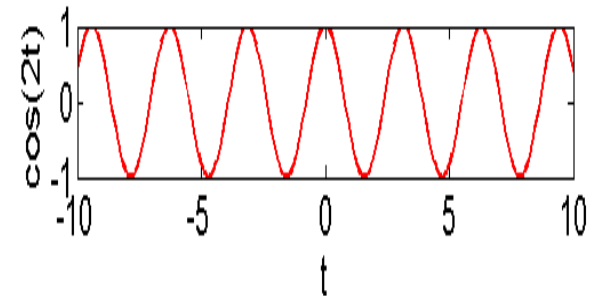
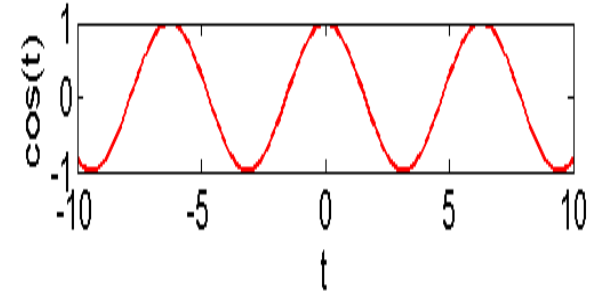
$$E_{\infty} = \infty$$

Average power:

$$P_{period} = \frac{1}{T_0} E_{period} = 1$$

Useful to consider **harmonic signals**

Terminology is consistent with its use in music, where each frequency is an integer multiple of a fundamental frequency



General Complex Exponential Signals

So far, considered the real and periodic complex exponential

Now consider when C can be complex. Let us express C in polar form and a in rectangular form:

$$C = |C|e^{j\phi}$$

$$a = r + j\omega_0$$

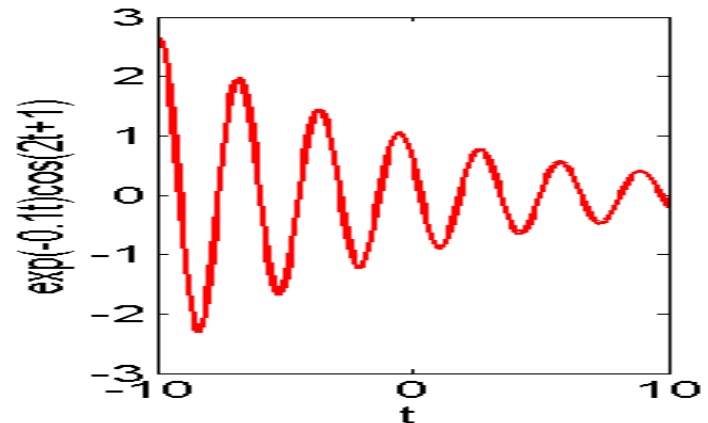
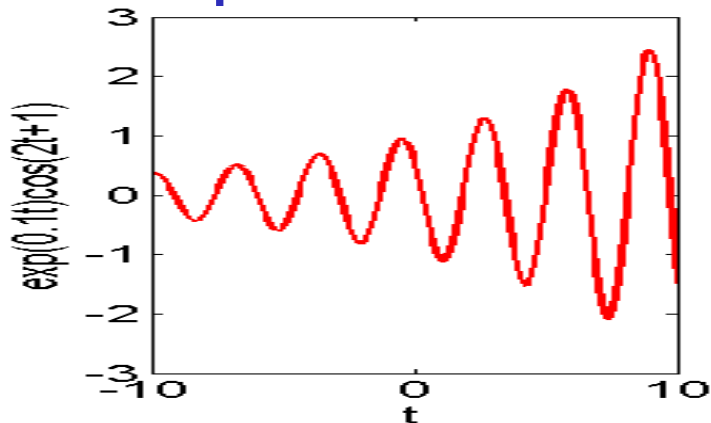
So

$$Ce^{at} = |C|e^{j\phi}e^{(r+j\omega_0)t} = |C|e^{rt}e^{j(\omega_0+\phi)t}$$

Using Euler's relation

$$Ce^{at} = |C|e^{j\phi}e^{(r+j\omega_0)t} = |C|e^{rt}\cos((\omega_0 + \phi)t) + j|C|e^{rt}\sin((\omega_0 + \phi)t)$$

These are **damped sinusoids**

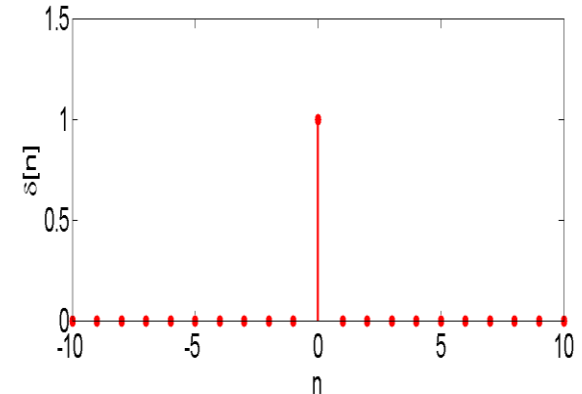


Discrete Unit Impulse and Step Signals

The discrete **unit impulse signal** is defined:

$$x[n] = \delta[n] = \begin{cases} 0 & n \neq 0 \\ 1 & n = 0 \end{cases}$$

Useful as a **basis** for analyzing other signals

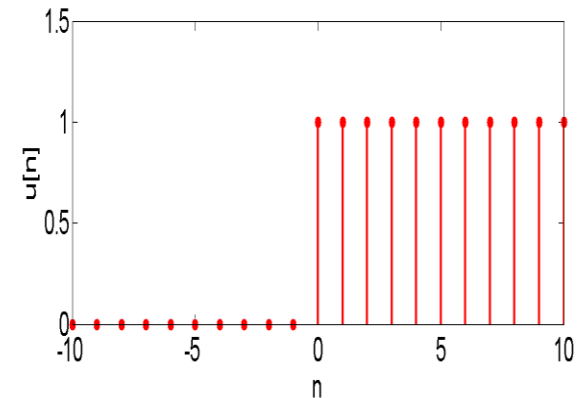


The discrete **unit step signal** is defined:

$$x[n] = u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}$$

Note that the unit impulse is the first difference (derivative) of the step signal

$$\delta[n] = u[n] - u[n-1]$$



Similarly, the unit step is the running sum (integral) of the unit impulse.

Continuous Unit Impulse and Step Signals

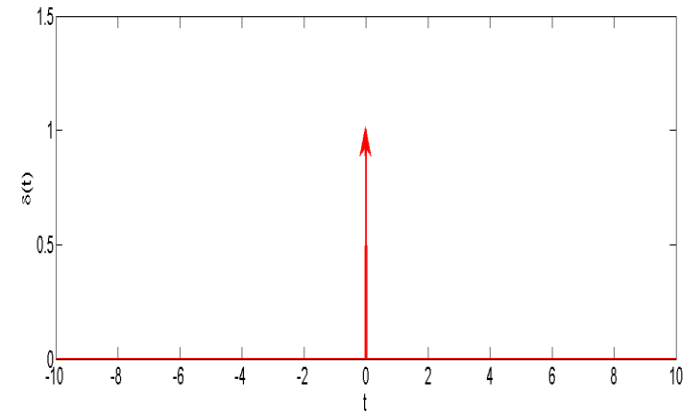
The continuous **unit impulse signal** is defined:

$$x(t) = \delta(t) = \begin{cases} 0 & t \neq 0 \\ \infty & t = 0 \end{cases}$$

Note that it is discontinuous at $t=0$

The arrow is used to denote area, rather than actual value

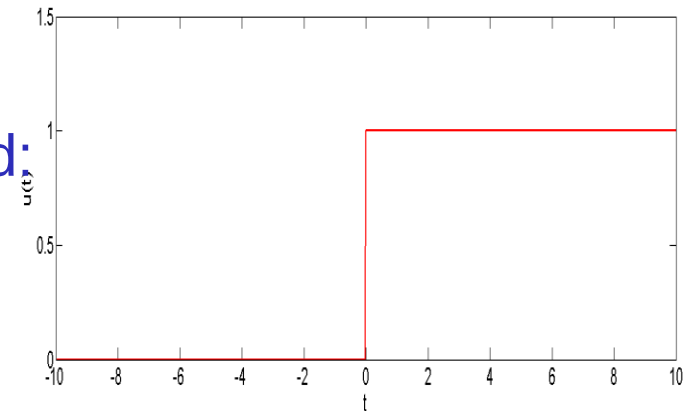
Again, useful for an infinite basis



The continuous **unit step signal** is defined:

$$x(t) = u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

$$x(t) = u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$



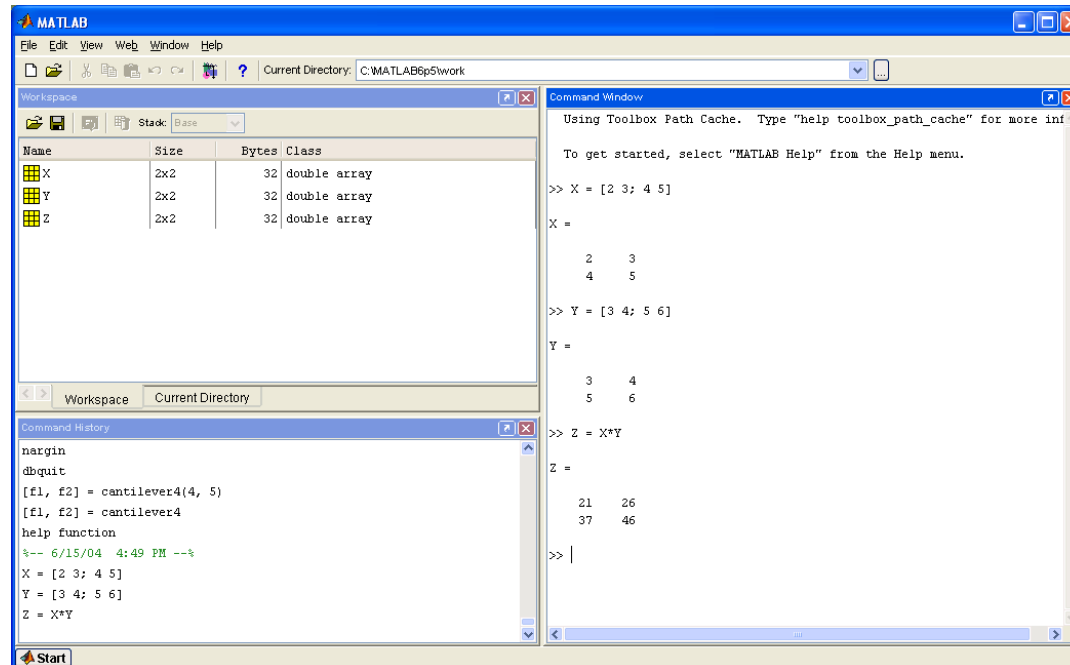
Introduction to Matlab

Simulink is a package that runs inside the Matlab environment.

Matlab (Matrix Laboratory) is a dynamic, interpreted, environment for matrix/vector analysis

User can build programs (in .m files or at command line) C/Java-like syntax

Ideal environment for programming and analysing discrete (indexed) signals and systems



Basic Matlab Operations

```
>> % This is a comment, it starts with a "%"
>> y = 5*3 + 2^2;           % simple arithmetic
>> x = [1 2 4 5 6];         % create the vector "x"
>> x1 = x.^2;               % square each element in x
>> E = sum(abs(x).^2);       % Calculate signal energy
>> P = E/length(x);         % Calculate av signal power
>> x2 = x(1:3);             % Select first 3 elements in x
>> z = 1+i;                 % Create a complex number
>> a = real(z);             % Pick off real part
>> b = imag(z);             % Pick off imaginary part
>> plot(x);                 % Plot the vector as a signal
>> t = 0:0.1:100;           % Generate sampled time
>> x3=exp(-t).*cos(t);      % Generate a discrete signal
>> plot(t, x3, 'x');        % Plot points
```

Other Matlab Programming Structures

Loops

```
for i=1:100
    sum = sum+i;
end
```

Goes round the for loop 100 times, starting at i=1 and finishing at i=100

```
i=1;
while i<=100
    sum = sum+i;
    i = i+1;
end
```

Similar, but uses a while loop instead of a for loop

Decisions

```
if i==5
    a = i*2;
else
    a = i*4;
end
```

Executes whichever branch is appropriate depending on test

```
switch i
case 5
    a = i*2;
otherwise
    a = i*4;
end
```

Similar, but uses a switch

Matlab Help!

These slides have provided a rapid introduction to Matlab

- Mastering Matlab 6, Prentice Hall,
- Introduction to Matlab (on-line)

Lots of help available

- Type `help` in the command window or `help operator`. This displays the help associated with the specified operator/function
- Type `lookfor topic` to search for Matlab commands that are related to the specified topic
- Type `helpdesk` in the command window or select help on the pull down menu. This allows you to access several, well-written programming tutorials.
- `comp.soft-sys.matlab` newsgroup

Learning to program (Matlab) is a “glued to seat” activity. There is no substitute for practice, making mistakes, understanding concepts

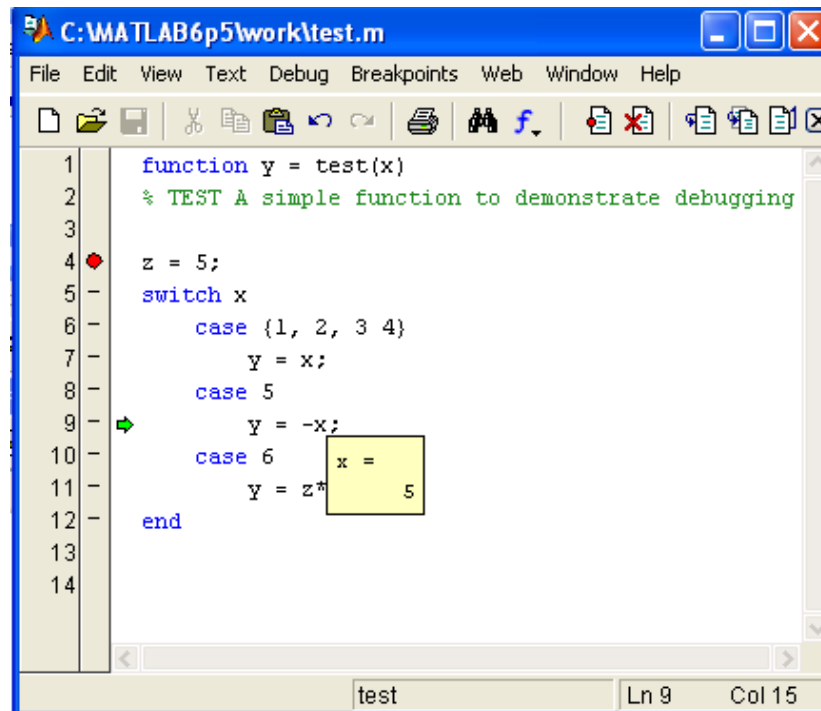
Using the Matlab Debugger

Because Matlab is an interpreted language, there is no compile type syntax checking and the likelihood of a run-time error is higher

Run-time debugging can help

Use the debug and breakpoints pull-down menus to determine where to stop program and inspect variables

Step over lines/step into functions to evaluate what happens



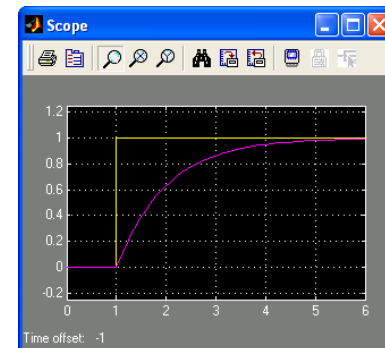
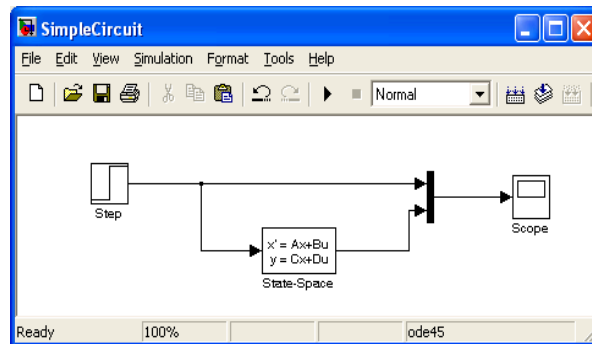
Introduction to Simulink

Simulink is a graphical, “drag and drop” environment for building simple and complex signal and system dynamic simulations.

It allows users to concentrate on the structure of the problem, rather than having to worry (too much) about a programming language.

The parameters of each signal and system block is configured by the user (right click on block)

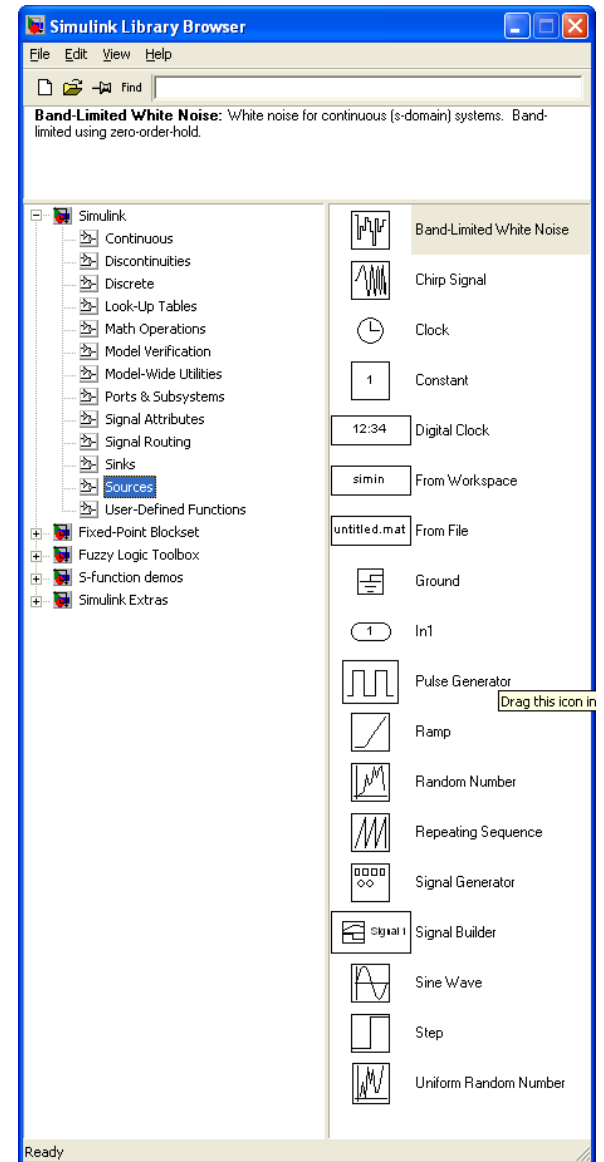
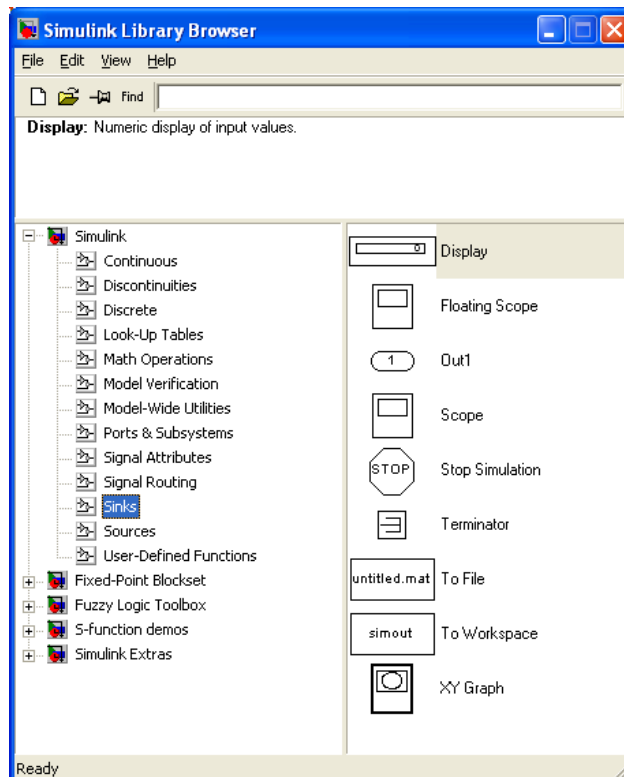
Signals and systems are simulated over a particular time.



Signals in Simulink

Two main libraries for manipulating signals in Simulink:

- **Sources:** generate a signal
- **Sink:** display, read or store a signal



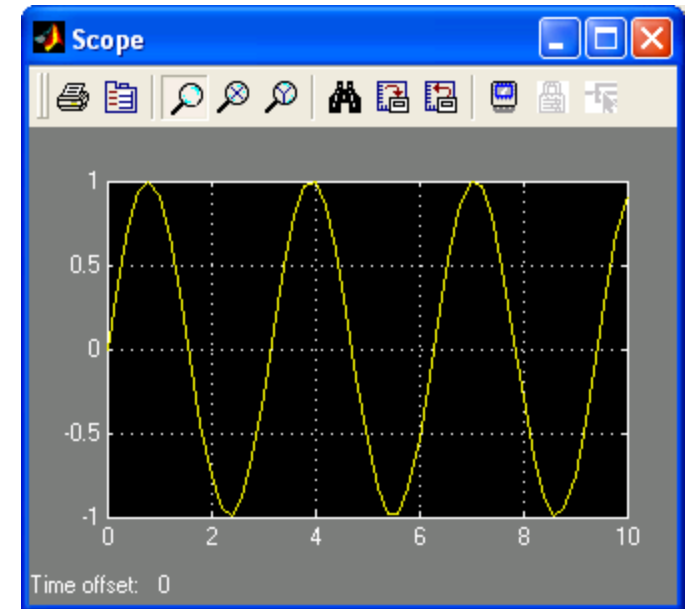
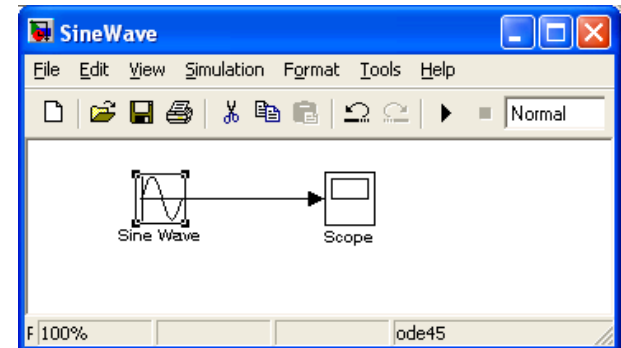
Example: Generate and View a Signal

Copy “sine wave” source and “scope” sink onto a new Simulink work space and connect.

Set sine wave parameters modify to 2 rad/sec

Run the simulation:
Simulation - Start

Open the scope and leave open while you change parameters (sin or simulation parameters) and re-run



Lecture 2: Summary

This lecture has looked at signals:

- Power and energy
- Signal transformations
 - Time shift
 - Periodic
 - Even and odd signals
- Exponential and sinusoidal signals
- Unit impulse and step functions

Matlab and Simulink are complementary environments for producing and analysing continuous and discrete signals.

This will require some effort to learn the programming syntax and style!

Lecture 2: Exercises

Text:

Q1.3

Q1.7-1.14

Matlab/Simulink

- Try out basic Matlab commands on slide 17
- Try creating the sin/scope Simulink simulation on slide 23 and modify the parameters of the sine wave and re-run the simulation
- Learning how to use the help facilities in Matlab is important - do it!